

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Student Project Repository and Showcase Platform

1. Introduction

1.1 Project Overview

The **Student Project Repository and Showcase Platform** is a web-based application that provides a centralized repository for storing, managing, approving, collaborating, and showcasing student projects. The platform enables students to submit **Internal Academic Projects**, **External Projects**, and **Project Ideas** while ensuring quality through faculty verification.

The system also introduces a **Credit Point System** to encourage students to actively contribute quality projects and collaborate with peers. Access to project repositories is controlled based on contribution level, ensuring that only active contributors can access valuable project resources.

2. Objectives

- Digitize project submission and approval.
 - Maintain separate repositories for Internal Projects, External Projects, and Project Ideas.
 - Automatically assign faculty reviewers for internal projects.
 - Allow students to select faculty guides for external projects.
 - Encourage innovation through Idea Showcase.
 - Reward active contributors using Credit Points.
 - Provide secure access control to project resources.
 - Encourage collaboration through project enhancement.
 - Display student rankings using a leaderboard.
 - Improve transparency in project management.
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3. Technology Stack

Component	Technology
Frontend	React.js
Backend	Node.js + Express.js
Database	PostgreSQL
Authentication	JWT
File Storage	Local Storage / Cloud Storage
Version Control	Git & GitHub

4. User Roles

Student

- Register/Login
 - Upload Internal Projects
 - Upload External Projects
 - Upload Project Ideas
 - View Repository
 - Enhance Projects
 - Earn Credit Points
 - View Leaderboard
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Faculty

- Review Internal Projects
 - Accept External Guide Requests
 - Review External Projects
 - Approve/Reject Projects
 - Award Credit Points
 - Provide Feedback
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Administrator

- View All Projects
- Remove Projects
- View Leaderboard

- Manage Users
 - Monitor Repository
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5. Functional Modules

Module 1 – User Authentication & Role Management

Description

Provides secure authentication and role-based access.

Features

- Student Registration
 - Faculty Login
 - Admin Login
 - Logout
 - Password Reset
 - JWT Authentication
 - Role-Based Dashboard
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Module 2 – Student Profile Management

Description

Maintains complete student information.

Features

- Personal Information
 - Department
 - Academic Year
 - Skills
 - Domain of Interest
 - Total Credit Points
 - Approved Project Count
 - Portfolio
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Module 3 – Internal Project Management

Description

Handles academic projects completed as part of the curriculum.

Submission Includes

- Project Title
- Abstract
- Documentation
- GitHub Repository
- PPT
- Demo Video (Optional)
- Vercel Link (Optional)

Workflow

Student Uploads Project

↓

System identifies project domain

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Faculty automatically assigned based on:

- Domain Expertise
- Current Availability

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Faculty Reviews

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Approved Project Published

Faculty approval is mandatory.

No guide request is required.

Module 4 – External Project Management

Description

Allows students to showcase projects completed outside academics.

Examples

- Hackathons
- Freelancing
- Personal Projects
- Open Source
- Competitions

Submission Includes

- Documentation
- GitHub Repository
- Certificate (Optional)
- Demo Video (Optional)
- Vercel Link (Optional)

Workflow

Student selects project domain

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System displays eligible faculty

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Student sends guide request

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Faculty accepts/rejects

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Student uploads project

↓

Faculty reviews

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Repository Publication

Faculty approval is mandatory.

Module 5 – Idea Showcase Module

Description

Allows students to publish project ideas before implementation.

Submission Includes

- Idea Title
- Documentation
- GitHub Repository (Optional)
- Demo Video (Optional)
- Vercel Link (Optional)

Features

- Direct Publication
 - No Faculty Approval Required
 - Students earn Idea Credit Points
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Module 6 – Faculty Assignment & Guide Management

Description

Manages faculty allocation.

Internal Projects

- Automatic faculty assignment
- Based on domain
- Based on faculty availability

External Projects

- Student selects preferred faculty
 - Faculty accepts/rejects guide request
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Module 7 – Credit Point Management

Description

Maintains the contribution score of every student.

Credit Allocation

Activity	Credits
Internal Project Approved	10
External Project Approved	20
Idea Published	5
Successful Project Enhancement (Owner)	10
10 Likes on Project	1

Credits are awarded by faculty after project approval.

Module 8 – Project Collaboration & Enhancement

Description

Allows students to extend or improve existing approved projects.

Workflow

Student requests project access

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Eligibility verified

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Project accessed

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Student enhances project

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Faculty verifies enhancement

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Project Owner receives Credit Points

The original owner receives credits for encouraging collaboration.

Module 9 – Repository Access Control

Description

Controls access to repositories based on student contribution.

Eligibility Criteria

Students must have at least **3 approved projects**.

Additionally:

Credit Points	Access Level
< 60	No Repository Access
≥ 60	View Idea Repository
≥ 100	View Internal Projects
≥ 200	View External Projects

Students who do not satisfy the above criteria cannot access project repositories.

Module 10 – Project Repository & Search

Features

Browse

- Ideas
- Internal Projects
- External Projects

Search By

- Project Title
- Student Name
- Department
- Domain
- Technology

Sort By

- Latest
 - Most Viewed
 - Highest Credits
 - Most Popular
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Module 11 – Student Leaderboard

Description

Displays rankings of students according to total Credit Points.

Visible To

- Students
- Faculty
- Administrator

Ranking Criteria

- Total Credit Points
- Number of Approved Projects

Filters

- Overall
 - Department
 - Academic Year
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Module 12 – Project Engagement Module

Description

Tracks project popularity similar to LinkedIn.

Features

- Likes
- Comments
- Views

Credit Policy

- Every 10 Likes = 1 Credit
- Comments do not generate credits

- Views do not generate credits

Engagement statistics improve project visibility.

Module 13 – Notification Module

Student Notifications

- Faculty Assigned
- Faculty Accepted Guide Request
- Project Approved
- Project Rejected
- Credit Points Awarded
- Collaboration Updates

Faculty Notifications

- New Guide Request
- New Project Submission
- Enhancement Verification

Admin Notifications

- New Project Uploaded
 - Reported Projects
 - Repository Updates
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Module 14 – Admin Management Module

Description

Provides complete administrative control over the repository.

Features

- View All Projects
- Remove Projects
- View Leaderboard
- View Student Profiles
- Manage Repository
- Manage Departments

- View Project Statistics
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6. Credit Point Policy

Activity	Credits
Internal Project Approved	10
External Project Approved	20
Idea Published	5
Project Enhancement (Owner Reward)	10
Every 10 Likes	1
Comments	0
Views	0

7. Non-Functional Requirements

- Responsive User Interface
 - Secure JWT Authentication
 - Role-Based Access Control
 - PostgreSQL Relational Database
 - Fast Search Performance
 - Secure File Upload
 - Reliable Backup and Recovery
 - Scalable Architecture
 - Cross-Browser Compatibility
 - Modular and Maintainable Code
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8. Database Tables

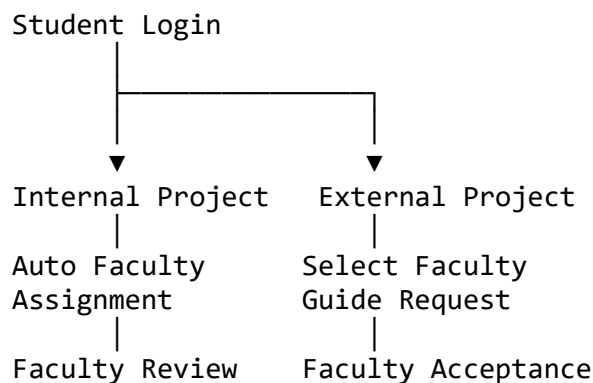
- Users
- Students
- Faculty
- Administrator
- Departments
- Faculty Availability
- External Guide Requests

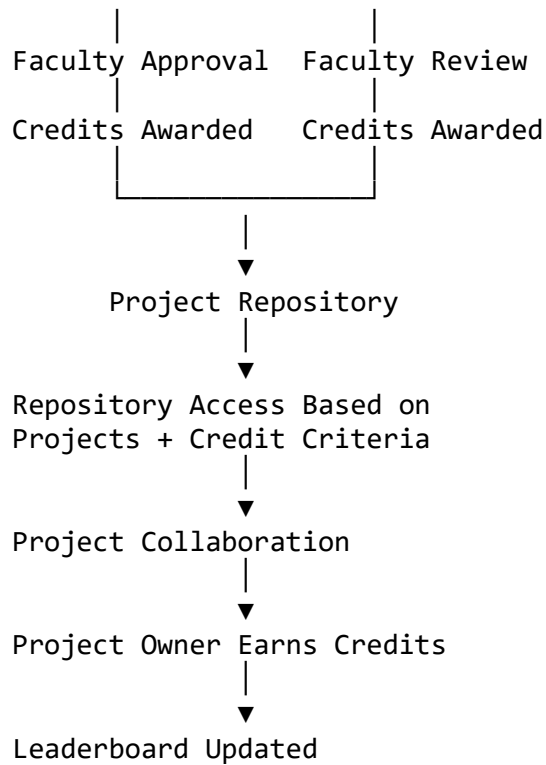
- Internal Projects
 - External Projects
 - Idea Showcase
 - Project Files
 - Project Reviews
 - Collaboration Requests
 - Credit Points
 - Leaderboard
 - Likes
 - Comments
 - Views
 - Notifications
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9. Expected Outcomes

- A centralized digital repository for student projects.
 - Transparent faculty-based project approval.
 - Automatic reviewer assignment for internal projects.
 - Faculty-guided external project showcase.
 - Promotion of innovative ideas through the Idea Showcase.
 - Fair contribution-based Credit Point system.
 - Secure access to repositories based on student contributions.
 - Increased collaboration through project enhancement.
 - Healthy competition through student leaderboards.
 - Better visibility of projects using likes, comments, and views.
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10. System Workflow





11. Future Enhancements

- Multi-college repository integration.
- Email notification service.
- QR code generation for project showcase.
- Project bookmarking.
- Version history for projects.
- Project plagiarism detection.
- Faculty workload analytics.
- Department-wise performance reports.
- Certificate generation for top contributors.
- Mobile application support.